

Network Packet Analysis Report Using Wireshark

Submitted By: Meghana Nagapuri

Internship Domain: Cybersecurity

Task: Packet Capture and Network Traffic Analysis

Tool Used: Wireshark

1. Objective

The objective of this task is to capture and analyze network packets using Wireshark in order to understand different network protocols, inspect packet communication, and identify unusual network behavior.

2. Tools and Environment

- Wireshark – Network packet analyzer
- Npcap – Packet capture driver
- Operating System – Windows 11
- Network Interface – Wi-Fi

3. Overview of Wireshark

Wireshark is a widely used network protocol analyzer that allows users to capture and inspect packets in real time. It helps cybersecurity professionals monitor network traffic, troubleshoot network problems, and analyze security incidents.

4. Procedure Followed

Step 1: Installed Wireshark and selected the active network interface.

Step 2: Started live packet capture while browsing websites and using internet services.

Step 3: Applied filters to analyze specific protocols and packet types.

Step 4: Observed packet communication and identified protocol behavior.

Step 5: Stopped capture and reviewed the collected traffic data.

5. Filters Used During Analysis

- dns – To analyze DNS traffic
- tls – To inspect encrypted HTTPS communication
- arp – To view ARP packets
- tcp.analysis.flags – To identify TCP anomalies
- tcp.flags.syn==1 && tcp.flags.ack==0 – To observe TCP SYN packets

6. Protocols Observed

DNS (Domain Name System)

Used for converting domain names into IP addresses.

TCP (Transmission Control Protocol)

Used for reliable communication and packet delivery.

TLS (Transport Layer Security)

Used for encrypted and secure communication over HTTPS.

ARP (Address Resolution Protocol)

Used for mapping IP addresses to MAC addresses.

7. Analysis Findings

- DNS requests and responses were successfully observed.
- TCP handshake communication was identified.
- HTTPS traffic encrypted using TLS was detected.
- ARP requests were present for local network communication.
- Minor retransmissions and duplicate acknowledgments were observed during traffic analysis.

8. Security Observations

The captured traffic mostly represented normal user browsing behavior. Minor TCP anomalies such as retransmissions and duplicate ACK packets were detected, which are commonly caused by network congestion or temporary connectivity issues. No malicious traffic or attacks were identified during analysis.

9. Recommendations

- Regularly monitor network traffic for suspicious activity.
- Use encrypted communication protocols such as HTTPS.
- Investigate excessive retransmissions if they occur frequently.
- Implement intrusion detection systems for continuous monitoring.
- Keep systems and network tools updated.

10. How to Perform This Task

Step 1: Install Wireshark on the system.

Step 2: Open Wireshark and choose the active network interface.

Step 3: Start packet capture and generate internet traffic by browsing websites.

Step 4: Apply protocol filters such as dns, tls, arp, and tcp.analysis.flags.

Step 5: Analyze captured packets and identify communication patterns.

Step 6: Save the capture file and prepare the analysis report.

11. Final Result

The network packet analysis task was successfully completed using Wireshark. Various network protocols including DNS, TCP, TLS, and ARP were analyzed. The captured traffic indicated normal network behavior, and no serious security threats were detected during the assessment.

12. Conclusion

This task provided practical exposure to network packet analysis and protocol inspection using Wireshark. Packet analysis plays an important role in cybersecurity by helping analysts monitor network communication, detect suspicious activity, and troubleshoot security issues.

13. Viva Questions and Answers

Q1. What is Wireshark?

Wireshark is a network protocol analyzer used to capture and inspect packets.

Q2. What is a packet?

A packet is a small unit of data transmitted across a network.

Q3. Why is TLS important?

TLS encrypts communication to provide secure data transfer.

Q4. What does DNS do?

DNS converts domain names into IP addresses.

Q5. What is packet analysis?

Packet analysis is the process of capturing and inspecting network traffic to understand communication behavior and identify security issues.

14. References

- Wireshark Official Documentation
- OWASP Network Security Guidelines
- Npcap Documentation